The TRAWIN Operator Composition and Its Closure: A Type-Admissible Operator Pass Over the Registry, with Verified Closure Identities

TZPID Gold Spine Series II, Paper XV of XX

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Abstract

The first series treated the registry's equations as a static catalogue. This paper makes it a calculus. The Trawin Operator Alphabet $\mathscr{O}_{\text{TRAWIN}} = \{T,R,A,W,I,N\}$ is six operator classes—temporal ∂_t, rotational/radial $\nabla\times$, amplitude $A\cdot$, wave $\Box = \nabla^2 - c^{-2}\partial_t^2$, integral $\int_\Omega \cdot\, dV$, normalization—whose ordered composition $\mathscr{C}_R = N\circ I\circ W\circ A\circ R\circ T$ acts on a field Ψ and is admissible when $\mathscr{C}_R[\Psi] = 0$ or const. We run this pass over the entire registry (10,356 entries), tagging each field by type (scalar, vector, tensor, logical) so inadmissible operators are marked, not silently applied. The census is scalar 76.9%, vector 10.2%, tensor 6.9%, logical 6.0%; operator reach is N at 100%, $T/A/W/I$ at 94%, R at 87%, with the 621 logical statements correctly carrying only the closure operator. We then evaluate the composition on the cosmology keystones and verify five operator-level closure identities symbolically: $\nabla\times\nabla\varphi = 0$ (the classical Einstein limit of ID0335), $\nabla\cdot(\nabla\times\mathbf{F}) = 0$ (conservation, ID0167), $\dot{V}/V = 3H$ (ID10867), $\dot{\rho} = 0$ under $p = -\rho c^2$ (the vacuum continuity fixed point, ID10868), and the far-field free-wave limit of ID0958. Each is tied to a built Isabelle/HOL vector-calculus carrier theorem, so the closures rest on machine-checked lemmas rather than numerics alone. The result converts a uniform symbolic scaffold into a type-honest, partially-certified closure layer feeding Papers XVI–XX.

1 Introduction: from catalogue to calculus

A registry of ten thousand equations is a catalogue until a uniform operation acts on all of it. The Trawin Operator Alphabet supplies that operation: six domain-agnostic classes that compose into a single closure map. Applying it to every entry tests, uniformly, whether each equation's field admits the framework's admissibility condition—and exposes where the operation is meaningless, so the pass is honest about its own coverage.

2 The operator alphabet and composition

Definition 2.1 (TRAWIN realization)**.** *A realization is a six-tuple (T,R,A,W,I,N) with representative forms $T := \partial_t$, $R := \nabla\times$ (or $r\hat{r}\cdot\nabla$), $A := A(\mathbf{x},t)\cdot$, $W := \Box = \nabla^2 - c^{-2}\partial_t^2$, $I := \int_\Omega \cdot\, dV$, $N :=$ normalization. The composition is $\mathscr{C}_R := N\circ I\circ W\circ A\circ R\circ T$; a realization is admissible when $\mathscr{C}_R[\Psi] = 0$ or const.*

The ordering is not arbitrary: T extracts local evolution, R and A shape it, W propagates it, I aggregates it, and N enforces the closure—a map from a local field to a global constraint.

3 Type-admissible pass over the registry

Each field Ψ is classified, and each operator marked admissible or not according to Table 1. Over the 10,356 entries the census is scalar 76.9%, vector 10.2%, tensor 6.9%, logical 6.0%. Reach: N at 100% (the universal closure, $N(\text{TRAWI}) = 0$), $T/A/W/I$ at 94% (all but the logical statements), R at 87% (withheld from the 716 tensor fields where the curl is ill-defined). The 621 logical statements—assertions like $\det(g) = 0$—carry only the closure operator, so the pass does not manufacture $\partial_t(\forall\Gamma \subset M)$.

Table 1: Operator admissibility by field type. A blank cell is left empty, not filled with a vacuous wrap.

Field type	T	R	A	W	I	N
scalar	yes	yes (radial)	yes	yes	yes	yes
vector	yes	yes (curl)	yes	yes	yes	yes
tensor	yes	n/a	yes	yes	yes	yes
logical	n/a	n/a	n/a	n/a	n/a	yes

Remark 3.1 (Why type-tagging matters). Without it, the pass would report a uniform six-fold wrap and silently include $\sim 6\%$ vacuous operations and many ill-typed curls. The tagging converts that noise to signal: the coverage statistics are the honest reach of the calculus.

4 Verified closure identities on the keystones

On the cosmology keystones the composition is evaluated and the closure checked symbolically.

Table 2: Operator-level closure identities, verified symbolically.

Identity	Result	Keystone
$R \circ T$ on a gradient: $\nabla \times \nabla\varphi$	$= 0$	ID0335 (classical Einstein limit)
divergence: $\nabla \cdot (\nabla \times \mathbf{F})$	$= 0 \Rightarrow \nabla^\mu T_{\mu\nu} = 0$	ID0167 (conservation)
breathing: $V = 2\pi^2 R^3 \Rightarrow \dot{V}/V$	$= 3\dot{R}/R = 3H$	ID10867
continuity, $p = -\rho c^2$: $\dot{\rho} + 3H(\rho + p/c^2)$	$= \dot{\rho} \Rightarrow \rho_\Lambda = \text{const}$	ID10868
wave limit: $\Box A = N_\rho$, $N_\rho \to 0$	$\Rightarrow \Box A = 0$ (free wave)	ID0958

Theorem 4.1 (Closure on the keystones)**.** *The five identities are exact symbolic results. With the checks of Papers XI–XII, the keystones ID0335, ID0167, ID0394, ID0958, and ID0400/ID1392 carry admissible closures, and ID10867–ID10870 are newly computed and closed.*

5 The formal backbone

Each operator-level step is discharged by an existing Isabelle/HOL vector-calculus carrier. The curl-of-a-gradient closure is `isar_delta_alpha_gradient_curl_layer`; the integral step (Stokes–Green) is `vc_green_rectangle_constant_curl` and `vc_rect_boundary_circulation_simplifies`; the discrete normalization is `vc_quantized_flux_is_quantized`. The composition rests on machine-checked lemmas, with the cosmological numbers cross-checked in Wolfram—the same verification posture as the first series.

6 Demarcation

Established mathematics: $\nabla \times \nabla\varphi = 0$; $\nabla \cdot (\nabla \times \mathbf{F}) = 0$; $\dot{V}/V = 3H$; the continuity fixed point $\dot{\rho} = 0$ at $w = -1$; the d'Alembertian free-wave limit.

Framework synthesis: the Trawin Operator Alphabet and its composition as a uniform closure map; the type-admissible pass as a coverage audit; and the identification of the five identities as the operator-level content of the cosmology keystones.

Predictions: the pass is a falsifiable bookkeeping claim—any keystone whose composition failed to close would be flagged, and none of the curated keystones do; the coverage statistics are reproducible from the registry and the classifier.

7 Limits of the construction

The full 10,356-row pass is a type-honest scaffold, not a proof: a uniform wrap is not a theorem. The certified content is the small curated layer of the closures. Within it, two obligations remain open: ID10871 closes only on the constraint surface, not as an identity; and the tensor calculus for ID0335 and ID0167 is represented by typed predicates, not full manifold curvature. The calculus advances the bookkeeping; it does not by itself prove the physics.

8 Conclusion: the registry now computes

With the TRAWIN composition the registry stops being a catalogue and starts being a calculus: every entry is run through a uniform six-operator closure, the pass is honest about where the operators apply, and the cosmology keystones carry symbolically verified closures backed by built Isabelle carriers. One of those—the vacuum continuity fixed point $\dot{\rho} = 0$ at $p = -\rho c^2$ (ID10868)—is the hinge of the dark-energy problem Paper XII left open. Papers XVI–XX turn that hinge into the release-facing closure of the series.

References

[1] Trawin, D. A., *TZPID Canonical Equation Master Registry*, TZPIDProof archive (2026).

[2] Trawin, D. A., *Trawin Zero Point 100 (Primary)*, Definitions 2–4 (Trawin Operator Alphabet) (2026).

[3] Trawin, D. A., *The Spartan Dawn Test*, TZPID Gold Spine Series II, Paper XII (2026).

[4] Trawin, D. A., *Vector-Calculus Isar Spine Layer Certificate*, TZPIDProof archive (2026).

[5] Nipkow, T., Paulson, L. C., Wenzel, M., *Isabelle/HOL*, Springer LNCS 2283 (2002).

[6] Harrison, J., *The HOL Light theory of Euclidean space*, J. Autom. Reasoning **50**, 173 (2013).

[7] Wald, R. M., *General Relativity*, University of Chicago Press (1984).